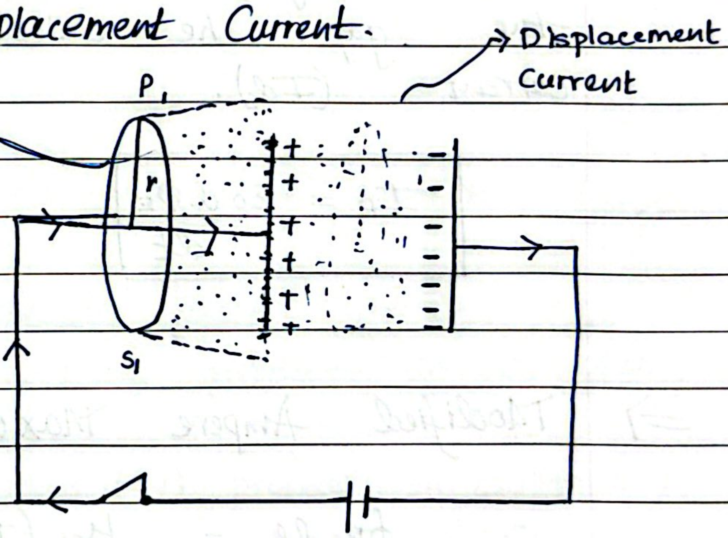
Chapter 8

Electromagnetic Waves.

→ Basic Idea of Displacement Current.

Ampere circuital Law applicable here
Here for a conducting current



Here consider a capacitor being charged by a battery through a wire take an amperian loop P_1, S_1

- Surface 1 (flat disc) cuts through (P_1 S_1) and conduction current flows through it

Amperes Law gives => $B \cdot dl = \mu_0 I$

- Surface 2 (bulged surface) here the (+ charged) capacitor plate charge stops here It cannot pass through thus. $\oint B \cdot dl = 0$

⇒ What happens here ?

Here the charge keeps flowing and thus as the flow of charge does not stop Electric field is produced between the two plates (in the gap).

Maxwell proposed that a changing electric field between the plates is itself equivalent to a current.
Even though no charge is passing through the gap, he called it Displacement Current (I_d).

$$\boxed{I_D = \varepsilon_0 \frac{d\phi_E}{dt}} \qquad \boxed{\mathcal{E} = \frac{d\phi_B}{dt}}$$

⇒ Modified Ampere Maxwell Law.

$$\oint B \cdot dl = \mu_0 (I_c + I_D)$$

$$\boxed{I_c = \frac{d\phi}{dt}} \quad \text{and} \quad \boxed{I_D = \varepsilon_0 \frac{d\phi_E}{dt}}$$

⇒ Properties of Displacement Current

- Caused : A time varying electric field / changing electric flux
- Where it exists : Between the plates of the capacitor.
- Produces magnetic field just like conduction current.
-

⇒ Relation between Conduction and Displacement Current.

$$I_D = \frac{\varepsilon_0 d\phi_E}{dt}$$

$$I_D = \varepsilon_0 \frac{d}{dt}(E \cdot A)$$

$(\sigma = q/A)$
$(\sigma A = q)$

$$I_D = \varepsilon_0 \frac{d}{dt}\left(\frac{\sigma}{\varepsilon_0} A\right)$$

$$I_D = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right)$$

$$\boxed{I_D = \frac{dq}{dt}} = I_C$$

⇒ Electromagnetic Waves.

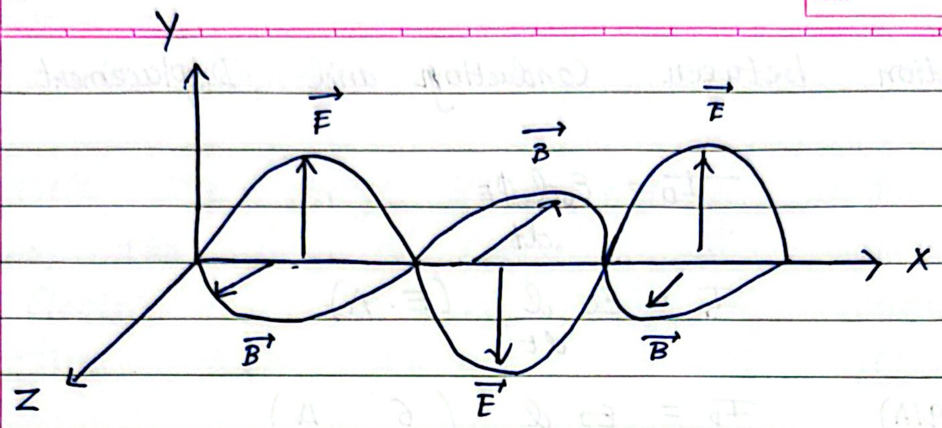
An accelerating (or oscillating) electric charge produces

- an oscillating electric field which (by Maxwell's Ampere Law) produces an oscillating magnetic field
- which (by faradays law) produces an oscillating E.F.
- and so on the two fields regenerate each other and propagate through space as a wave without medium.

electric field — magnetic field — $\boxed{\frac{E}{B} = c}$ (speed of light

$$C = \frac{1}{\sqrt{\mu_0 \mu_r \varepsilon_0 K}}$$

Note: E.M waves consist of oscillating electric field (E) and magnetic field (B). mutually perpendicular



Both E and B vary sinusoidally and are exactly in phase with each other

⇒ Important Characteristics of Electromagnetic Waves

i) The electromagnetic waves are produced by accelerated charge.
ii) These waves do not require any material medium for propagation.
iii) These waves travel in free space with speed of light ($3 \times 10^8\ ms^{-1}$) $\boxed{c = 1/\sqrt{\mu_0 \varepsilon_0}}$
iv) The energy in electromagnetic waves is divided equally between electric and magnetic fields $\boxed{u_e = u_m}$

⇒ Transverse Nature of Electromagnetic Waves

Significance : This transverse nature allows polarisation for eg: light passing through polarising sunglasses.

⇒ Electromagnetic Spectrum.

Sr.no	Name	Production	Properties	Uses
1.	Gamma Rays $\lambda(1\times10^{-4}$ to $1\times10^{-12} m)$ $f(3\times10^{22}$ to $3\times10^{18} Hz)$	By transitions of atomic nuclei and decay of certain elements particles.	Chemical reaction on photographic plates, fluoroscence ionisation diffraction highly penetrating harmful to body.	Provide info about structure of atomic nuclei sterilize medical equipmants, to kill cancer cells and radiotherapy
2.	X – Rays $\lambda(10^{-13}$ to $3\times10^{-8} m)$ $f(3\times10^{21}$ to $1\times10^{16} Hz)$	By sudden decelaration of high speed e's at high atomic number target and also by electronic transition among orbit of atoms	All properties of gamma rays but less penetrating	For studying structures of inner atomic electron shells for detection of fractures bones and healing bones for detecting holes in engineered products
3.	Ultraviolet Radiation. $\lambda(6\times10^{-10}$ to $4\times10^{-7} m)$ $f(3\times10^{17}$ to $7\times10^{14} Hz)$	By sun, arc vaccum spark and ionised gases	All properties of gamma rays but less penetrating produce photo electric effect harmful to body.	In dettection of invisible writing forged documents fingerprints and to preserve food to destroy bacteria.
4.	Visible Light $\lambda(4\times10^{-7}$ to $7\times10^{-7} m)$ $f(7\times10^{14}$ to $4\times10^{14} Hz)$	Radiated by excited atoms in gases or incandescent bodies	Reflection, refraction interference, diffraction, polarisation photographic action sensation of sight	Reveals structure of molecules and arrangement of electrons in external shells of atom

Sr.no	Name	Production	Properties	Uses.
5.	Infrared Radiation $\lambda(8\times10^{-7}$ to $5\times10^{-3}m)$ $f(4\times10^{14}$ to $6\times10^{11}Hz)$	From hot bodies and by rotational and vibrational transitions in molecules	Heating effect on thermopile bolometer, reflection, refraction and penetration through fog.	For providing electrical energy to satellites by means of solar cells, weather forecasting and in warfare to look through fog.
6.	Microwaves. $\lambda(1\times10^{-3}$ to $3\times10^{-1}m)$ $f(3\times10^{11}$ to $1\times10^{9}Hz)$	By oscillating currents in special vaccum tubes and by electromagnetic oscillators	Reflection, polarisation.	In radar system used in aircraft navigation long distance communication via satelites and in oven
7.	Radio waves $\lambda(1\times10^{-1}$ to $1\times10^{4}m)$ $f(3\times10^{9}$ to $3\times10^{4}Hz)$	By oscillating electrical circuits	Reflection, diffraction	In radio and TV communication systems cellular phones use radio waves to transmit in VHF band.
8.	Long waves $\lambda(5\times10^{6}$ to $6\times10^{6}m)$ $f(60$ to 50 Hz$)$	Weak radiation from A.C circuits.		